APPENDIX A

THE CHEAT SHEET

Plain words first, formula second.

WHAT YOU WANT	PLAIN WORDS	FORMULA
Workload	Leverage times reps times sets times tempo multiplier	$V = L \cdot R \cdot S \cdot M_{\text{TUT}}$
Tempo multiplier	Seconds per rep divided by 2, capped at 2	$M_{\text{TUT}} = \min\left(\frac{t}{2},\ 2\right)$
Reps in a hold	Seconds held divided by 2	$R = \frac{t_{\text{hold}}}{2}$
Density	Workload per minute of the session	$\frac{V}{\text{minutes}}$
Working reps	Your max clean reps minus 2	$R_w = R_{\max} - 2$
Max reps at the next rung	About three quarters of your max now	$R_{\text{next}} \approx 0.75\, R_{\text{now}}$
Spike ratio	This week's V over your 4-week average	$\frac{V_{\text{week}}}{\bar{V}_4}$
Highest safe week	1.10 times your 4-week average	$V_{\max} = 1.10 \times \bar{V}_4$
Session budget (sets)	12 minus 2 times your work load points	$B = 12 - 2W$
Sleep debt	Hours short of 7, added over 7 nights	$\sum \max(0,\ 7 - h)$
Session minutes	Sets times (work plus rest) over 60, plus prep	$\frac{S(w + r)}{60} + p$
Compliance	Workouts with all five gates over all workouts	$\frac{N_{\text{gates}}}{N_{\text{all}}} \times 100$
Plan efficiency	V done over V planned	$\frac{V_{\text{done}}}{V_{\text{planned}}}$
Consistency	Workouts done over workouts planned	$\frac{N_{\text{done}}}{N_{\text{planned}}} \times 100$
Deload week	60% of your peak V	$V_{\text{deload}} = 0.60 \times V_{\text{peak}}$

WHAT YOU WANT	PLAIN WORDS	FORMULA
Weeks to rebuild	Restart fraction f, rebuilt at 10% a week	$\frac{\ln(1/f)}{\ln(1.10)}$
Cost per workout	Total cost over workouts done	$\frac{\text{Cost}}{N_{\text{done}}}$
Chance of a false alarm	One minus the chance all n numbers stay quiet	$P = 1 - (1 - p)^n$

The leverage ladders. L is a rounded planning number, not a lab value. Your log is the authority.

PATTERN	RUNG	MOVE	L
Push	1	Wall push-up	0.30
	2	Counter push-up (waist-high hands)	0.45
	3	Bench push-up (knee-high hands)	0.55
	4	Floor push-up	0.65
	5	Feet-elevated push-up	0.75
Pull	1	Steep inverted row (body near upright)	0.35
	2	Inverted row, knees bent	0.50
	3	Inverted row, body straight	0.65
	4	Band-assisted pull-up	0.80
	5	Pull-up	1.00
Hinge	1	Glute bridge	0.30
	2	Hip hinge, hands on hips	0.40
	3	Single-leg bridge	0.50
	4	Single-leg Romanian deadlift (bodyweight)	0.70
	5	Nordic curl, lowering only (assisted)	0.90
Squat	1	Box squat, high box	0.40
	2	Bodyweight squat	0.60
	3	Split squat	0.70
	4	Rear-foot-elevated split squat	0.85
	5	Pistol squat to a box	1.00
Core	1	Dead bug	0.30
	2	Bird dog	0.40
	3	Front plank	0.50
	4	Side plank	0.60
	5	Hanging knee raise	0.80

APPENDIX B

JOINT LONGEVITY PROTOCOL TABLE

General education, not a diagnosis. Pain that is new, severe, or that worries you deserves a professional.

The 5-minute prep block, in order:

1. **Wrists (40 seconds):** wrist circles, then 10 slow rocks on hands and knees.
2. **Shoulders (60 seconds):** 10 wall slides, then 10 small arm circles.
3. **Spine (60 seconds):** 8 cat-camels, then 5 dead bugs each side.
4. **Hips (80 seconds):** 8 hip circles each way, then 8 slow 90/90 switches.
5. **Ankles (60 seconds):** 10 ankle rocks each side, then 10 slow calf raises.

JOINT	DAILY PREP (ABOUT 1 MINUTE)	BUILDER (2 TIMES A WEEK)	GET CHECKED IF
Wrists	Circles; slow rocks on hands and knees	Plank on hands, shifting weight forward and back, 20 to 30 s	Numbness or tingling that stays; swelling; pain on any weight-bearing
Elbows	Open and close fists; slow bends and straightens	Slow 4-second lowering on rows and push-ups; 20 s hold mid-range	Sharp pain that worsens each session; locking; swelling
Shoulders	Wall slides; small arm circles	Scapular push-ups and hanging shoulder shrugs, 2 × 8	Night pain; arm weakness; catching or giving way
Neck and upper back	Chin tucks; extension over a chair back	Prone Y-T-W raises on the floor, 2 × 8 each	Pain with arm numbness; dizziness or headache with training
Lower back	Cat-camel; dead bug	Bird dog, side plank, and a small curl-up, 2 sets each	Leg numbness or weakness; loss of bladder or bowel control (emergency); pain after a fall
Hips	Hip circles; 90/90 switches	Single-leg bridge; split squat with a pause at the bottom	Groin pain with clicking or locking; night pain
Knees	Slow box squat; wall tibialis raises	Box squat with 3-second lowering; wall sit 30 to 45 s	Swelling; locking; giving way; pain on stairs that keeps worsening
Ankles and feet	Ankle rocks; slow calf raises	Single-leg calf raise, 3-second lowering; one-leg balance 30 s	Cannot bear weight after a twist; sharp heel or Achilles pain that worsens

APPENDIX C

THE FIVE MOVEMENT WORDS

Every workout in this book uses these five ideas and no others.

1. **Push:** you move your body away from your hands, or your hands away from your body. Push-ups, dips, and overhead pressing are push moves. At work: shoving a cart, opening a heavy door.
2. **Pull:** you move your body toward your hands, or your hands toward your body. Rows and pull-ups are pull moves. At work: hauling, climbing, dragging.
3. **Hinge:** you bend at the hip with a steady spine. Bridges, hinges, and deadlifts are hinge moves. At work: lifting a box off the floor.
4. **Squat:** you bend at the knee and hip together, getting low and standing back up. Box squats, split squats, and pistols are squat moves. At work: working low to the ground, standing up from a chair.
5. **Core:** you keep the spine steady while the arms and legs do something else. Planks, dead bugs, and side planks are core moves. At work: carrying a heavy load across a floor.

Clean rep: full range, controlled tempo, steady spine, no sagging, no twisting, and no pain above 2. IF a rep is not clean, THEN it does not count.

Status: these describe a working model. The manual says movement patterns and a ladder of rungs organise training well for most people. It does not say they fit every body, every injury, or every goal. Your own tested log is the only authority.

APPENDIX D

WORKOUT JOURNAL TEMPLATE

Copy one block per workout. Four are printed here. Photocopy this page for more.

SESSION #	DATE	START TIME
_____	_____	_____

WORK LOAD TODAY (W)	HOURS SLEPT LAST NIGHT
0 / 1 / 2 / 3	_____

THE FIVE GATES

1. READY: SLEEP DEBT (HOURS)	PAIN (0 TO 10)	RED FLAG?
_____	_____	Y / N

2. WARM: WRIST	SHOULDER	SQUAT	BALANCE
P / F	P / F	P / F	P / F

3. PLAN: SESSION BUDGET (SETS)	PLANNED V	SPIKE RATIO
_____	_____	_____ *(must be 1.10 or less)*

4. RUNG: CHOSEN FROM MY LOG, NOT FROM PRIDE?
Y / N

5. STOP RULE: WRITTEN ON THE PAGE?
Y / N

THE WORK

PUSH: MOVE AND L	REPS X SETS	SECONDS PER REP	V
PULL: MOVE AND L	REPS X SETS	SECONDS PER REP	V
HINGE: MOVE AND L	REPS X SETS	SECONDS PER REP	V
SQUAT: MOVE AND L	REPS X SETS	SECONDS PER REP	V
CORE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

THE RESULT

SESSION V	MINUTES	DENSITY (V / MINUTES)
PAIN AFTER (0 TO 10)	NEXT-MORNING PAIN	PLAN EFFICIENCY (DONE / PLANNED)

THE AUDIT

1. ALL FIVE GATES BEFORE THE FIRST SET?
Y / N

2. EVERY COUNTED REP CLEAN?
Y / N

3. STOPPED EACH SET 2 CLEAN REPS SHORT?
Y / N

4. PAIN 2 OR LESS, SETTLED IN 24 HOURS?
Y / N

5. V, PAIN, AND SLEEP LOGGED?
Y / N

6. EFFORT SCORE (1 TO 5)

_____

COMPLIANCE
PASS / FAIL

ONE SENTENCE: WHAT WOULD I DO DIFFERENTLY?

_____

_____

SESSION #	DATE	START TIME

WORK LOAD TODAY (W) 0 / 1 / 2 / 3	HOURS SLEPT LAST NIGHT

THE FIVE GATES

1. READY: SLEEP DEBT (HOURS)	PAIN (0 TO 10)	RED FLAG? Y / N

2. WARM: WRIST P / F	SHOULDER P / F	SQUAT P / F	BALANCE P / F

3. PLAN: SESSION BUDGET (SETS)	PLANNED V	SPIKE RATIO *(must be 1.10 or less)*

4. RUNG: CHOSEN FROM MY LOG, NOT FROM PRIDE? Y / N

5. STOP RULE: WRITTEN ON THE PAGE? Y / N

THE WORK

PUSH: MOVE AND L	REPS X SETS	SECONDS PER REP	V

PULL: MOVE AND L	REPS X SETS	SECONDS PER REP	V

HINGE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

SQUAT: MOVE AND L	REPS X SETS	SECONDS PER REP	V

CORE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

THE RESULT

SESSION V	MINUTES	DENSITY (V / MINUTES)
PAIN AFTER (0 TO 10)	NEXT-MORNING PAIN	PLAN EFFICIENCY (DONE / PLANNED)

THE AUDIT

1. ALL FIVE GATES BEFORE THE FIRST SET? Y / N
2. EVERY COUNTED REP CLEAN? Y / N
3. STOPPED EACH SET 2 CLEAN REPS SHORT? Y / N
4. PAIN 2 OR LESS, SETTLED IN 24 HOURS? Y / N
5. V, PAIN, AND SLEEP LOGGED? Y / N
6. EFFORT SCORE (1 TO 5)
COMPLIANCE PASS / FAIL

ONE SENTENCE: WHAT WOULD I DO DIFFERENTLY?

SESSION #	DATE	START TIME

WORK LOAD TODAY (W)	HOURS SLEPT LAST NIGHT
0 / 1 / 2 / 3	

THE FIVE GATES

1. READY: SLEEP DEBT (HOURS)	PAIN (0 TO 10)	RED FLAG?
		Y / N

2. WARM: WRIST	SHOULDER	SQUAT	BALANCE
P / F	P / F	P / F	P / F

3. PLAN: SESSION BUDGET (SETS)	PLANNED V	SPIKE RATIO
		(must be 1.10 or less)

4. RUNG: CHOSEN FROM MY LOG, NOT FROM PRIDE?
Y / N

5. STOP RULE: WRITTEN ON THE PAGE?
Y / N

THE WORK

PUSH: MOVE AND L	REPS X SETS	SECONDS PER REP	V

PULL: MOVE AND L	REPS X SETS	SECONDS PER REP	V

HINGE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

SQUAT: MOVE AND L	REPS X SETS	SECONDS PER REP	V

CORE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

THE RESULT

SESSION V	MINUTES	DENSITY (V / MINUTES)
PAIN AFTER (0 TO 10)	NEXT-MORNING PAIN	PLAN EFFICIENCY (DONE / PLANNED)

THE AUDIT

1. ALL FIVE GATES BEFORE THE FIRST SET? Y / N
2. EVERY COUNTED REP CLEAN? Y / N
3. STOPPED EACH SET 2 CLEAN REPS SHORT? Y / N
4. PAIN 2 OR LESS, SETTLED IN 24 HOURS? Y / N
5. V, PAIN, AND SLEEP LOGGED? Y / N
6. EFFORT SCORE (1 TO 5)
COMPLIANCE PASS / FAIL

ONE SENTENCE: WHAT WOULD I DO DIFFERENTLY?

SESSION #	DATE	START TIME

WORK LOAD TODAY (W) 0 / 1 / 2 / 3	HOURS SLEPT LAST NIGHT

THE FIVE GATES

1. READY: SLEEP DEBT (HOURS)	PAIN (0 TO 10)	RED FLAG? Y / N

2. WARM: WRIST P / F	SHOULDER P / F	SQUAT P / F	BALANCE P / F

3. PLAN: SESSION BUDGET (SETS)	PLANNED V	SPIKE RATIO *(must be 1.10 or less)*

4. RUNG: CHOSEN FROM MY LOG, NOT FROM PRIDE?
Y / N

5. STOP RULE: WRITTEN ON THE PAGE?
Y / N

THE WORK

PUSH: MOVE AND L	REPS X SETS	SECONDS PER REP	V

PULL: MOVE AND L	REPS X SETS	SECONDS PER REP	V

HINGE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

SQUAT: MOVE AND L	REPS X SETS	SECONDS PER REP	V

CORE: MOVE AND L	REPS X SETS	SECONDS PER REP	V

THE RESULT

SESSION V	MINUTES	DENSITY (V / MINUTES)
PAIN AFTER (0 TO 10)	NEXT-MORNING PAIN	PLAN EFFICIENCY (DONE / PLANNED)

THE AUDIT

1. ALL FIVE GATES BEFORE THE FIRST SET? Y / N
2. EVERY COUNTED REP CLEAN? Y / N
3. STOPPED EACH SET 2 CLEAN REPS SHORT? Y / N
4. PAIN 2 OR LESS, SETTLED IN 24 HOURS? Y / N
5. V, PAIN, AND SLEEP LOGGED? Y / N
6. EFFORT SCORE (1 TO 5)
COMPLIANCE PASS / FAIL

ONE SENTENCE: WHAT WOULD I DO DIFFERENTLY?

APPENDIX E

50-WORKOUT LOG

#	DATE	SLEEP DEBT (H)	V DONE	PAIN (0 TO 10)	ALL 5 GATES? (Y/N)	SPIKE RATIO
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
31						

#	DATE	SLEEP DEBT (H)	V DONE	PAIN (0 TO 10)	ALL 5 GATES? (Y/N)	SPIKE RATIO
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						

After 50 workouts, work out:

WHAT	HOW	RESULT
Consistency	Workouts done, divided by workouts planned	
Compliance	Workouts with all five gates, divided by 50	
Average weekly V	Add up V done, divide by the number of weeks	
Highest spike ratio	The biggest number in the last column	
Rungs climbed	Count them for each pattern	
Average next-morning pain	Add up next-morning pain, divide by 50	
Longest layoff	The longest run of weeks with no workouts	

E-2. WEEKLY LEDGER (32 WEEKS). One line per week. Spike ratio is this week's V divided by the average of the last four weeks. Mark every fourth week as a deload.

WEEK	DONE / PLANNED	WEEKLY V	AVG OF LAST 4	SPIKE RATIO	DELOAD?
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
26					
27					
28					
29					
30					
31					

WEEK	DONE / PLANNED	WEEKLY V	AVG OF LAST 4	SPIKE RATIO	DELOAD?
32					

E-3. RUNG TEST LOG. Test a rung only when the five gates are open. Working reps are max clean reps minus 2.

DATE	PATTERN	RUNG	MAX CLEAN REPS	WORKING REPS	MOVE UP? (Y/N)

DATE	PATTERN	RUNG	MAX CLEAN REPS	WORKING REPS	MOVE UP? (Y/N)

E-4. PAIN LOG. Score pain from 0 to 10. 0 to 2 is green, 3 to 4 is amber, 5 or more is red. A next-morning rise of 2 or more means go down one rung.

DATE	WHERE	DURING	AFTER	NEXT MORNING	LIGHT (G/A/R)	ACTION TAKEN

DATE	WHERE	DURING	AFTER	NEXT MORNING	LIGHT (G/A/R)	ACTION TAKEN

E-5. SLEEP AND JOB-LOAD SHEET. Write the hours slept each night. Sleep debt is the sum of (7 minus hours), counting only nights under 7.

WEEK OF	MON	TUE	WED	THU	FRI	SAT	SUN	DEBT (H)	W (0 TO 3)

E-6. TWELVE-WEEK BLOCK PLANNER. Set the target V before the week starts. Never plan more than 1.10 times the average of the last four weeks. Week 4, 8, and 12 are deloads at 60 percent of peak.

WEEK	PLANNED WORKOUTS	TARGET V	V DONE	PAIN PEAK	NOTES
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					

E-7. MOBILITY CHECK LOG. Pass or fail each check. M is the number of passes (0 to 4).

DATE	WRIST	SHOULDER	SQUAT	BALANCE	M (0 TO 4)	WHAT I DID ABOUT IT

DATE	WRIST	SHOULDER	SQUAT	BALANCE	M (0 TO 4)	WHAT I DID ABOUT IT

E-8. SESSION PLAN SHEET. Fill this in at gate 3, before the first set. Write the rung from your log, not from pride.

DATE	PUSH	PULL	HINGE	SQUAT	CORE	PLANNED V

DATE	PUSH	PULL	HINGE	SQUAT	CORE	PLANNED V

E-9. MONTHLY REVIEW. Once a month, add up the log. Change one thing only.

MONTH	WORKOUTS DONE	COMPLIANCE	AVG WEEKLY V	TOP SPIKE RATIO	AVG PAIN	ONE CHANGE

E-10. REBUILD SHEET. After a layoff, restart at the fraction f from the restart table. Raise V by no more than 1.10 times per week.

WEEK	FRACTION F	TARGET V	V DONE	SPIKE RATIO	PAIN PEAK
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

E-11. FIELD NOTES. One line per lesson. Date it. Keep it short.

DATE	WHAT I LEARNED

E-12. MY OWN RULES. Write your stop rules, your red lines, and the rungs you will not skip. Sign and date it.

RULE	DATE